

Analysis Process — OLAP

Project Architecture

Enterprise OLAP platform integrating transactional data sources with batch analytics, BI dashboards, and exportable reports.

Built with Apache Spark, Apache Airflow, Apache Superset, PostgreSQL, MongoDB, Hive, Neo4j, and Elasticsearch.

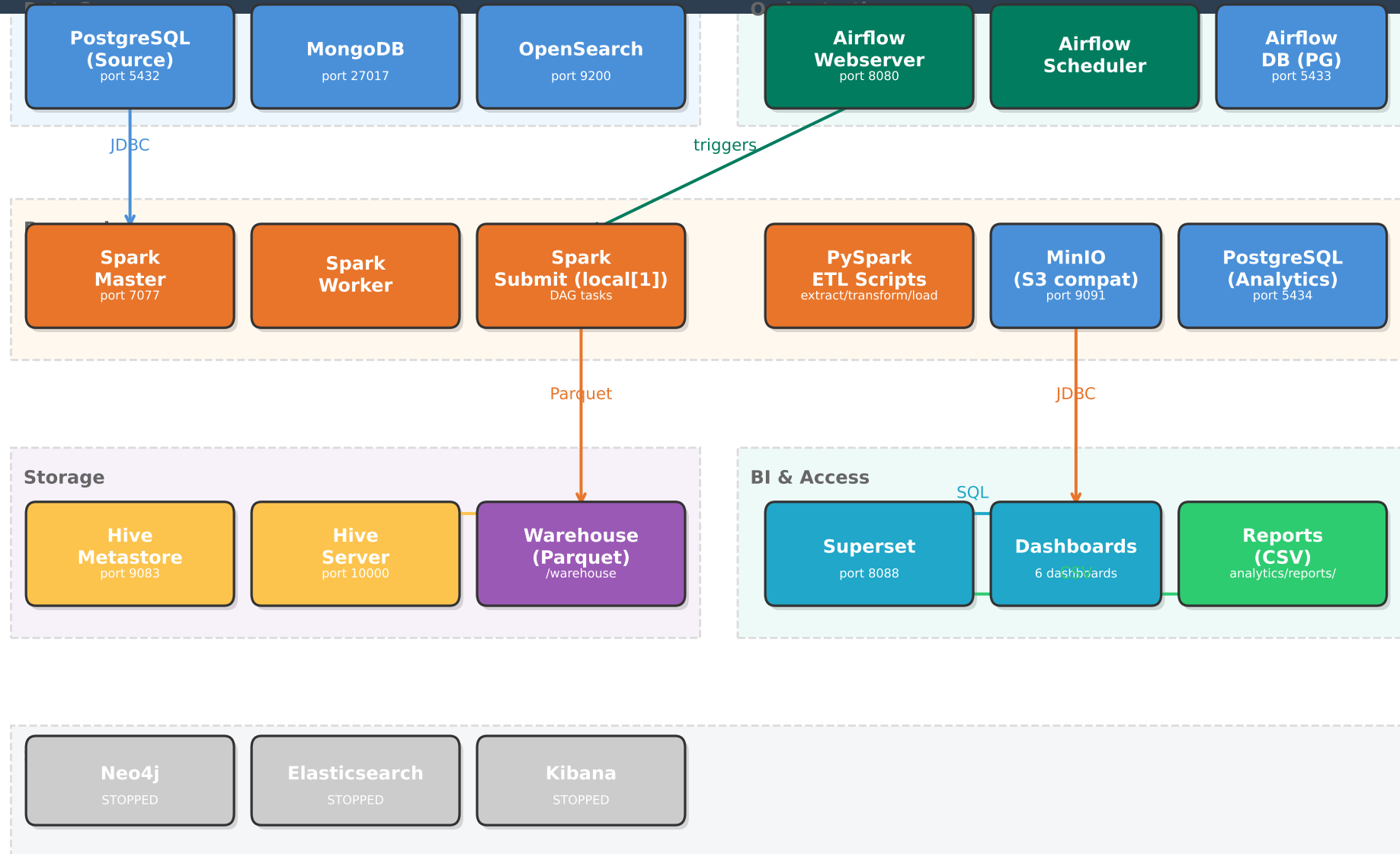
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Environment: Docker Compose (25 containers)

Repository: Analysis-Process_OLAP



Architecture



ne — DAG Flow

EXTRACT

1

extract.py

Spark → JDBC
Read PostgreSQL
Partitioned read
Staging parquet

TRANSFORM

2

transform.py

Spark SQL
Aggregations
Analytics joins
CSV intermediates

LOAD

3

load.py

Spark → Parquet
Write to /warehouse
Validation checks
Integrity report

INIT SUPERSET

4

init_superset.py

Superset REST API
Import data sources
Configure database

CREATE DASHBOARDS

5

create_dashboards.py

Execution Context

- Superset REST API
- Orchestrate with Airflow 2.10
- Create datasets
- Build dashboards
- Processing: Spark local[1] (512m)
- 6 dashboards
- Schedule: Daily at 02:00 UTC
- Language: Python 3.10 / PySpark
- Status: Pipeline completed OK

EXPORT REPORTS

6

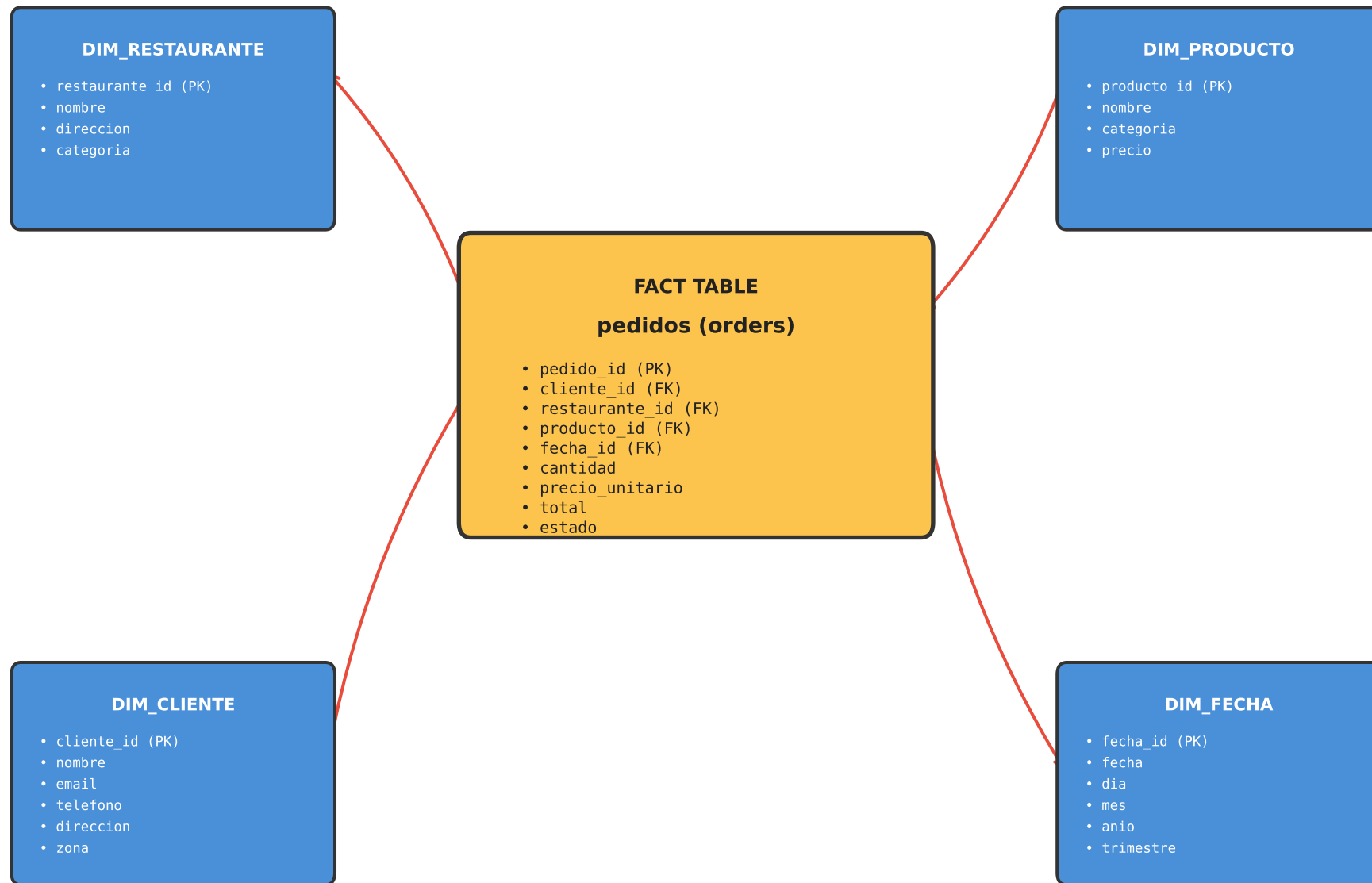
export_reports_pg.py

PostgreSQL SQL
Aggregate queries
Write CSVs
15 report files

Generated Reports (15 files)

- actividad_clientes_zona.csv
- ingresos_por_mes_categoria.csv
- pedidos_completados_vs_cancelados.csv
- productos_mas_vendidos_categoria.csv
- productos_mas_vendidos.csv
- promedio_gasto_cliente.csv
- restaurantes_mas_populares.csv
- ventas_por_dia_hora.csv
- (7 more in analytics/reports/)

— Star Schema



FK = Foreign Key | PK = Primary Key | 1 fact + 4 dimensions

← → One-to-Many relationship

ology Stack

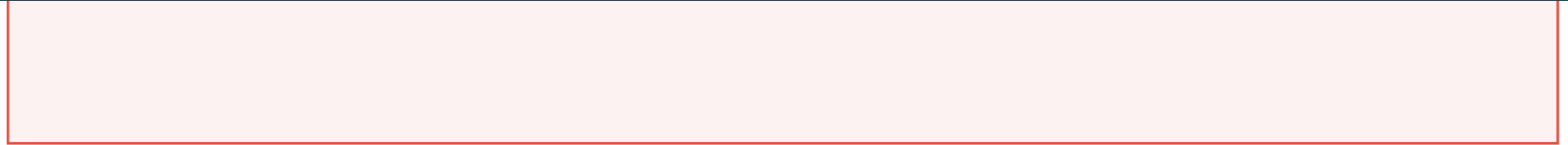
Component	Version	Purpose	Status
Apache Spark	3.5.8	Distributed data processing (ETL)	Operational
Apache Airflow	2.10.4	Workflow orchestration & scheduling	Operational
Apache Superset	4.1.1	BI dashboards & data visualization	Operational
PostgreSQL (Src)	16	Source transactional database	Operational
PostgreSQL (Ana)	16	Analytics database for Superset	Operational
PostgreSQL (Air)	16	Airflow metadata database	Operational
Apache Hive	4.0.1	Data warehouse (metastore + server)	Operational
MongoDB	7.0	Document store (source data)	Operational
MinIO	latest	S3-compatible object storage	Operational
OpenSearch	2.18	Search & indexing engine	Operational
Neo4j	4.4.48	Graph database (relationships + routes)	Operational
Elasticsearch	8.17	Full-text search engine	STOPPED
Kibana	8.17	Elasticsearch visualization	STOPPED
Docker	lastest	Containerization platform	Operational
Python	3.10	Scripting & PySpark	Operational

s & Future Work

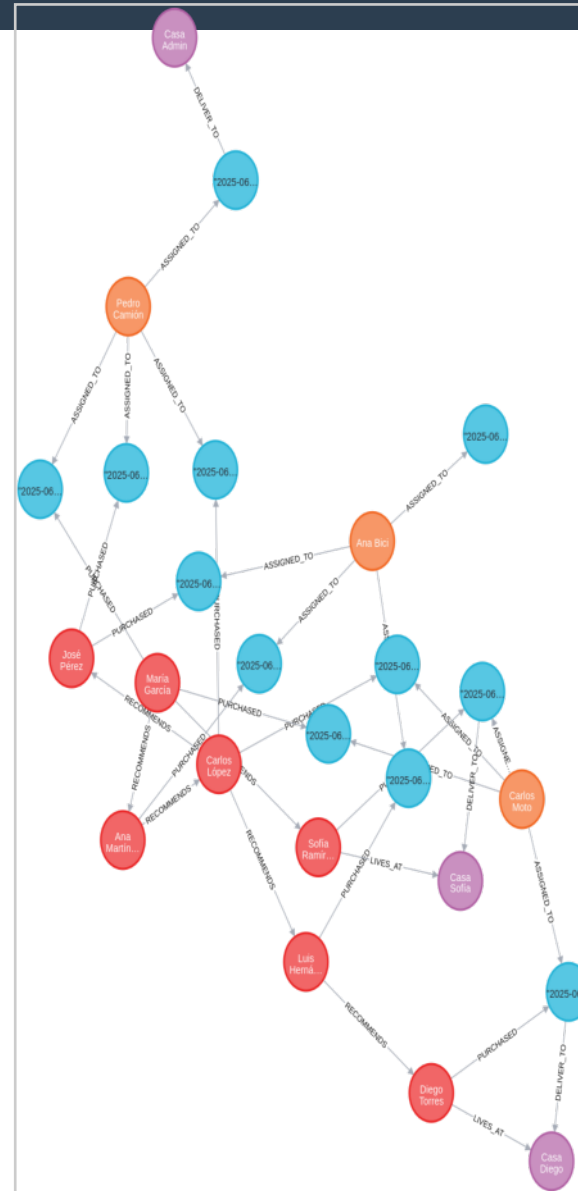
5. Superset Dashboard Drift

LOW — dashboards are static; schema changes require manual sync

Dashboards are created programmatically via REST API. If the underlying schema changes, dashboards may break silently.



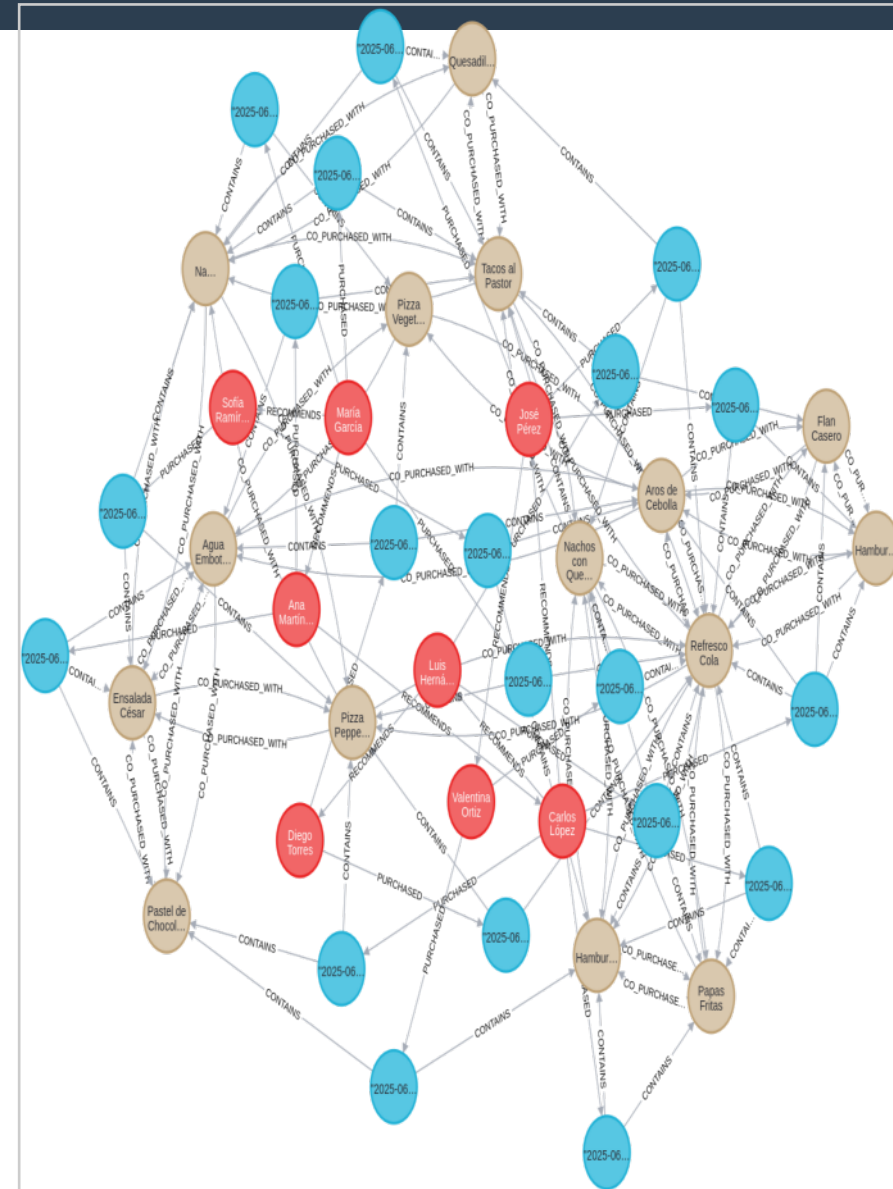
– Todos los nodos



todosNodos

Todos los nodos — Vista general del grafo completo (usuarios, productos, pedidos, ubicaciones, restaurantes, couriers)

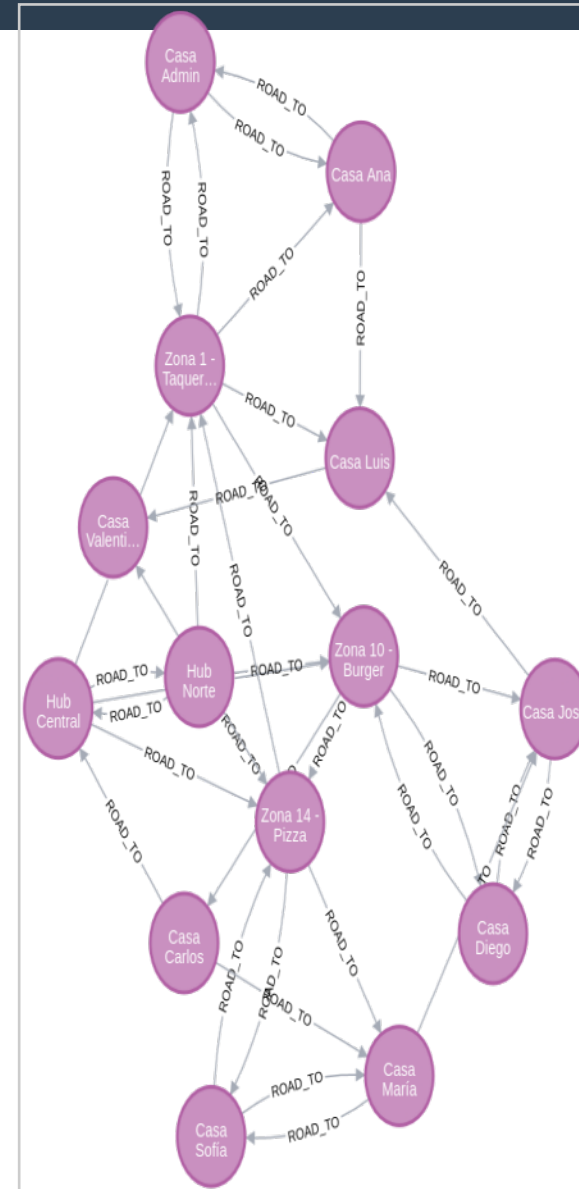
ios, pedidos y productos



usuariosPedidosYProductos

Usuarios, pedidos y productos — Relaciones PURCHASED y CONTAINS entre clientes, pedidos y productos

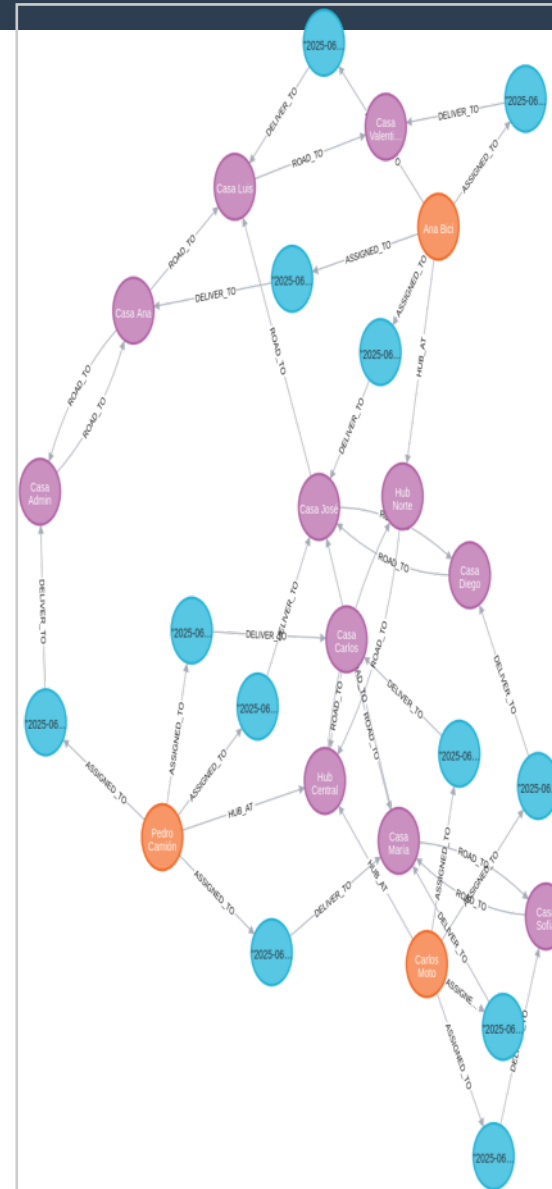
Red vial con ubicaciones



redVialConUbicaciones

Red vial con ubicaciones — Conexiones ROAD_TO entre hubs, restaurantes y domicilios con distancias y tiempos

er hubs y rutas de entrega



cuurierHubsYRutasDeEntrega

Courier hubs y rutas de entrega — Asignación de pedidos a couriers con rutas optimizadas nearest-neighbour